Siddharth Sondhi

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Portfolio: SiddharthSondhi.github.io

Education

- **San Jose State University** — San Jose, CA
  Master of Science in Computer Science; GPA: 3.81 — *Aug. 2024 – May. 2026*
- **UC Riverside** — Riverside, CA
  Bachelor of Science in Computer Science; GPA: 3.65 — *Sep. 2020 – Jun. 2024*

Projects

- **OpenGL Graphics Pipeline Renderer**: Developed a real time renderer in C++ using OpenGL and GLSL shaders. Implemented features including Phong lighting, model loading, transparency blending, cube map sky boxes, framebuffer based post processing effects, and an ImGUI interface.
- **AI Assisted VR Anatomy Education App**: Developed a Unity based virtual reality anatomy environment integrating LLM natural language interaction with traditional VR UI controls. Integrated local LLM inference, speech recognition, and function calling pipelines, followed by a comparative user study between AI assisted and traditional VR interaction methods.
- **N-Body Galaxy Simulation with Octree**: Simulated large scale gravitational interactions of thousands of particles using Euler integration and optimized performance through an octree based Barnes-Hut algorithm. Implemented using C++ and Open Frameworks.
- **Ray Marcher**: Developed a ray marcher to render 3D shapes defined by signed distance functions. Implemented using C++ and Open Frameworks.
- **Ray Tracer**: Developed a recursive ray tracer to simulate realistic light with reflections, refractions, and shadows. Implemented Phong shading for material appearance and handled multiple bounces of light to produce images. Implemented using C++.

Experience

- **Verve Group**
  Internship — *2023 Summer*
  - **Role**: Contributed to a data science project by extracting and analyzing device and location data to derive user insights for targeted advertising. Investigated correlations between demographic data (e.g., income brackets from US Census) and click behavior.
  - **Languages & Tools**: Python, Pandas, Git, command line, JupyterLabs
- **Inmobi**
  Internship — *2021 Summer*
  - **Role**: Automated a previously manual process of uploading certain data through creating a new tab on a web application. Used javascript and HTML to create the tab, parse the files, implement error checking, and implement user feedback. Once the file was uploaded, Python was used to make API calls and upload the parsed data to the company's database.
  - **Languages & Tools**: Javascript, Python/Django, Git, command line, HTML/CSS, Docker

Skills

- **Programming**: C++, Python, Java
- **Other Tools**: Git/Github, command line, HTML
- **Graphics and Simulation**: OpenGL, GLSL, Ray Tracing, Ray Marching, Particle Systems
- **Spoken Languages**: Fluent in English and Hindi